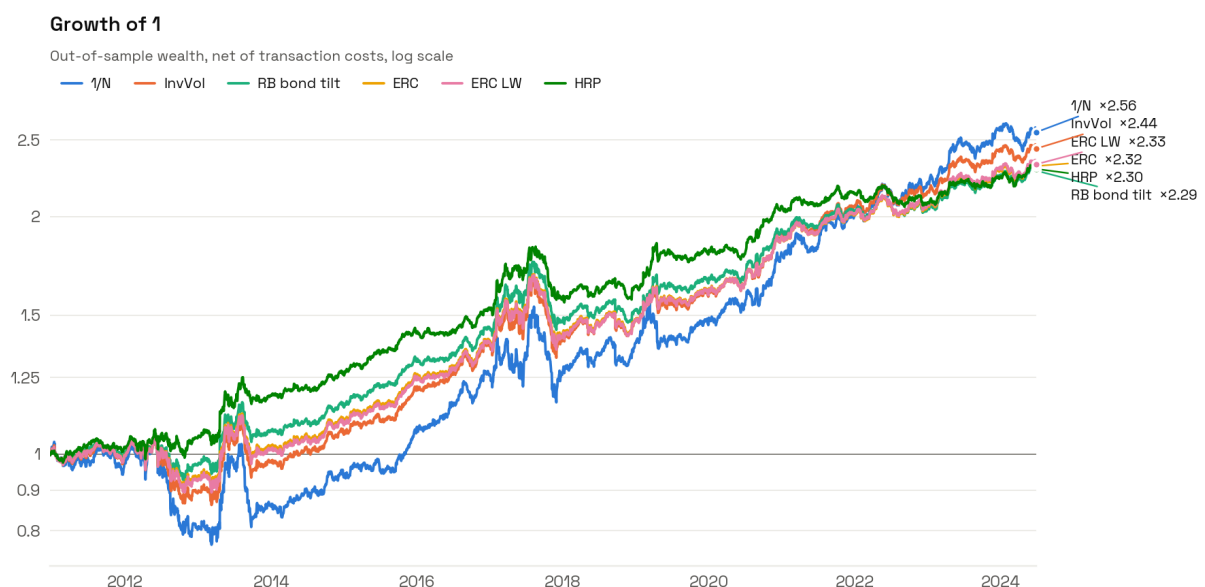


Risk-based portfolio allocation

A guide to the portopt library:
definitions, results and proofs

portopt compares risk-based allocation rules (1/N, inverse volatility, risk budgeting, ERC, HRP) in a walk-forward backtest, then measures their performance and risk. This guide first introduces the library, then covers each building block with the same structure: definition, result, proof, call in the code. Every stated result is either proved or flagged as admitted, with its reference.



Author Jean Philippe N'DRI
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Code portopt/

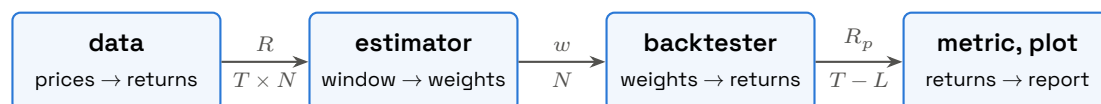
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1 Getting started

1.1 What the library does

portopt answers one question: *which allocation rule would have delivered the best return/risk trade-off, without using any future information?* It follows a four-step flow, one module per step.



1. **data** downloads prices, cleans them and computes the returns R , a `DataFrame` of T dates by N assets (section 2).
2. **estimator** turns a window of returns into weights w : this is the *allocation rule* (sections 3 to 5).
3. **backtester** applies the rule at every rebalancing date, lets the portfolio evolve between dates and returns its out-of-sample returns R_p (section 6).
4. **metric** and **plot** summarize these returns into metrics and charts (sections 7 to 9).

1.2 First backtest

```
pip install -e ".[dev]" # numpy, pandas, scipy, yfinance, matplotlib, pytest
```

```

from portopt import load_returns, EqualWeight, ERC, HRP, run_backtests, summary
from portopt import plot

# 1. data: daily returns, T dates x N assets
R = load_returns(["SPY", "TLT", "GLD", "DBC", "EFA"], start="2008-01-01")

# 2. estimator: the allocation rules to compare
rules = {"1/N": EqualWeight(), "ERC": ERC(), "HRP": HRP()}

# 3. backtester: 252-day window, rebalancing every 21 days
rets, bts = run_backtests(R, rules, in_sample=252, out_of_sample=21,
                          transaction_cost_bps=5)

# 4. metric and plot
print(summary(rets))
plot.save_report(rets, bts, "outputs/charts")

```

`rets` holds one daily out-of-sample return per strategy, and `bts["ERC"].weights_` the ERC weights at each rebalancing. Without network access, the `simulate_returns()` function of the `data` module provides a simulated universe of 10 ETFs.

1.3 Writing your own rule

A rule inherits from `BaseEstimator` and implements `_compute_weights`. The backtester accepts it as is.

```

import numpy as np
from portopt import BaseEstimator

class InverseVariance(BaseEstimator):

```

```
def _compute_weights(self, returns):
    cov = self._cov(returns)          # applies cov_method
    return 1.0 / np.diag(cov)
```

`fit` does the rest: it rejects windows with missing values, sets negative weights to zero, normalizes the sum to 1 and stores the result in `weights_`, a `pd.Series` indexed by asset.

1.4 API map

Object	Role	Section
<code>load_returns</code> , <code>download_prices</code>	get prices and returns	2
<code>clean_prices</code> , <code>compute_returns</code>	clean data, compute returns	2
<code>simulate_returns</code>	simulated universe for offline work	2
<code>estimate_cov</code> , <code>ledoit_wolf_cov</code> , <code>ewma_cov</code>	estimate the covariance	3
<code>risk_contributions</code>	risk contributions	4
<code>EqualWeight</code> , <code>InverseVolatility</code>	rules without optimization	5.1, 5.2
<code>RiskParity</code> , <code>ERC</code>	risk budgeting, equal contributions	5.3, 5.4
<code>HRP</code>	Hierarchical Risk Parity	5.5
<code>Backtester</code> , <code>run_backtests</code>	walk-forward backtest	6
<code>summary</code> , <code>metric module</code>	performance and risk metrics	7–9
<code>plot.save_report</code>	chart report (PNG)	10

1.5 Notation

N, T	number of assets, number of dates
$R_t \in \mathbb{R}^N$	simple returns of the assets, from the close of $t - 1$ to the close of t
Δ_N	$\{w \in \mathbb{R}^N : w \geq 0, \mathbf{1}^\top w = 1\}$, the set of long-only portfolios
$\Sigma, \sigma_i, \rho_{ij}$	covariance, volatilities $\sigma_i = \sqrt{\Sigma_{ii}}$, correlations
P	number of periods per year (252 for daily data)
L, H	estimation window length, holding period
$\bar{R}, \hat{\sigma}$	sample mean and standard deviation
Φ, φ	cumulative distribution function and density of the standard normal

2 Returns and portfolio

Definition 2.1 (returns). For a price P_t adjusted for dividends and splits, the simple return and the log return are

$$R_t = \frac{P_t}{P_{t-1}} - 1, \quad r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t).$$

Definition 2.2 (portfolio). A portfolio is a vector $w \in \Delta_N$: w_i is the share of value invested in asset i . Its return is denoted $R_{p,t}$.

Proposition 2.3. If w are the weights at $t - 1$, then $R_{p,t} = w^\top R_t$. The analogous identity is false for log returns.

Proof. Let n_i be the number of units of asset i held and $V_t = \sum_i n_i P_{i,t}$. Then $V_t/V_{t-1} = \sum_i \frac{n_i P_{i,t-1}}{V_{t-1}} \cdot \frac{P_{i,t}}{P_{i,t-1}} = \sum_i w_i (1 + R_{i,t})$, hence $R_{p,t} = w^\top R_t$. For log returns, $r_{p,t} = \ln \sum_i w_i e^{r_{i,t}}$, which is strictly greater than $\sum_i w_i r_{i,t}$ as soon as the $r_{i,t}$ differ (strict concavity of the logarithm). ■

The whole library therefore works with **simple returns**.

Definition 2.4 (portfolio volatility). If Σ is the covariance of returns, the variance and volatility of portfolio w are $\sigma_p^2(w) = w^\top \Sigma w$ and $\sigma_p(w) = \sqrt{w^\top \Sigma w}$.

Definition 2.5 (annualization). For returns with P periods per year: $\mu_{\text{ann}} = P \bar{R}$ and $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$.

Proposition 2.6. *If the returns of the P periods of a year are independent, with mean μ and variance σ^2 , and the annual return is their sum, then it has mean $P\mu$ and standard deviation $\sqrt{P} \sigma$.*

Proof. Expectation is linear, and by independence the variance of a sum is the sum of the variances: $\text{Var}(\sum_{k=1}^P X_k) = P\sigma^2$. ■

Additivity is exact for log returns and holds to first order for daily simple returns. This is what justifies the factors P and $\sqrt{P}$.

In the code

`load_returns(tickers, start, end, cache_dir="data_cache")` downloads adjusted prices, cleans them and returns simple returns. Cleaning (`clean_prices`) fills gaps of at most 5 days with the last known value, never fills with a future value, and keeps the missing values of an asset not yet listed: the backtester only invests in it once its history is complete.

3 Estimating the covariance

Every rule except 1/N uses an estimate of Σ over the window.

Definition 3.1 (sample covariance). Over a window of T dates, with $x_t = R_t - \bar{R}$: $S = \frac{1}{T-1} \sum_{t=1}^T x_t x_t^\top$.

S is unbiased but noisy. It has $N(N+1)/2$ parameters and, if $T \leq N$, it has rank at most $T-1$, so it cannot be inverted.

Theorem 3.2 (Marchenko-Pastur). *If the R_t are independent with covariance I_N and $N, T \rightarrow \infty$ with $N/T \rightarrow q \in (0, 1]$, the eigenvalues of S spread over $[(1 - \sqrt{q})^2, (1 + \sqrt{q})^2]$. (admitted, [Marchenko and Pastur, 1967](#))*

Even when all true eigenvalues equal 1, those of S spread from 0.09 to 2.91 for $q = 0.5$. Small eigenvalues are underestimated, and they are precisely the ones that inverting Σ amplifies.

Definition 3.3 (Ledoit-Wolf). Let $S_T = \frac{1}{T} \sum_t x_t x_t^\top$, $\|A\|^2 = \text{tr}(AA^\top)/N$ and $\mu = \text{tr}(S_T)/N$. Define

$$d^2 = \|S_T - \mu I\|^2, \quad \bar{b}^2 = \frac{1}{T^2} \sum_{t=1}^T \|x_t x_t^\top - S_T\|^2, \quad \delta^* = \frac{\min(\bar{b}^2, d^2)}{d^2} \in [0, 1], \quad \hat{\Sigma} = \delta^* \mu I + (1 - \delta^*) S_T.$$

d^2 measures the distance from S_T to the target μI , $\bar{b}^2$ the estimation noise of S_T ; δ^* is the share of the distance attributed to noise. This choice asymptotically minimizes the quadratic error among combinations of S_T and μI (admitted, [Ledoit and Wolf, 2004](#)).

Proposition 3.4. *Let $\lambda_1 \geq \dots \geq \lambda_N$ be the eigenvalues of S_T and $\delta^* > 0$.*

- (i) $\hat{\Sigma}$ has the same eigenvectors as S_T , with eigenvalues $\delta^* \mu + (1 - \delta^*) \lambda_j$, and it is positive definite.

(ii) Its condition number $\lambda_{\max}/\lambda_{\min}$ is lower than that of S_T .

Proof. (i) If $S_T v = \lambda v$, then $\hat{\Sigma} v = (\delta^* \mu + (1 - \delta^*) \lambda) v$, and these values are at least $\delta^* \mu > 0$. (ii) With $a = \delta^* \mu > 0$ and $c = 1 - \delta^*$, the inequality $\frac{a+c\lambda_1}{a+c\lambda_N} \leq \frac{\lambda_1}{\lambda_N}$ is equivalent to $a\lambda_N \leq a\lambda_1$, which holds. ■

Lemma 3.5 (computing $\bar{b}^2$ without forming the $x_t x_t^\top$). $\sum_t \|x_t x_t^\top - S_T\|_F^2 = \sum_t (x_t^\top x_t)^2 - T \|S_T\|_F^2$.

Proof. $\|x_t x_t^\top - S_T\|_F^2 = (x_t^\top x_t)^2 - 2x_t^\top S_T x_t + \|S_T\|_F^2$, and $\sum_t x_t^\top S_T x_t = \text{tr}(S_T \sum_t x_t x_t^\top) = T \|S_T\|_F^2$. ■

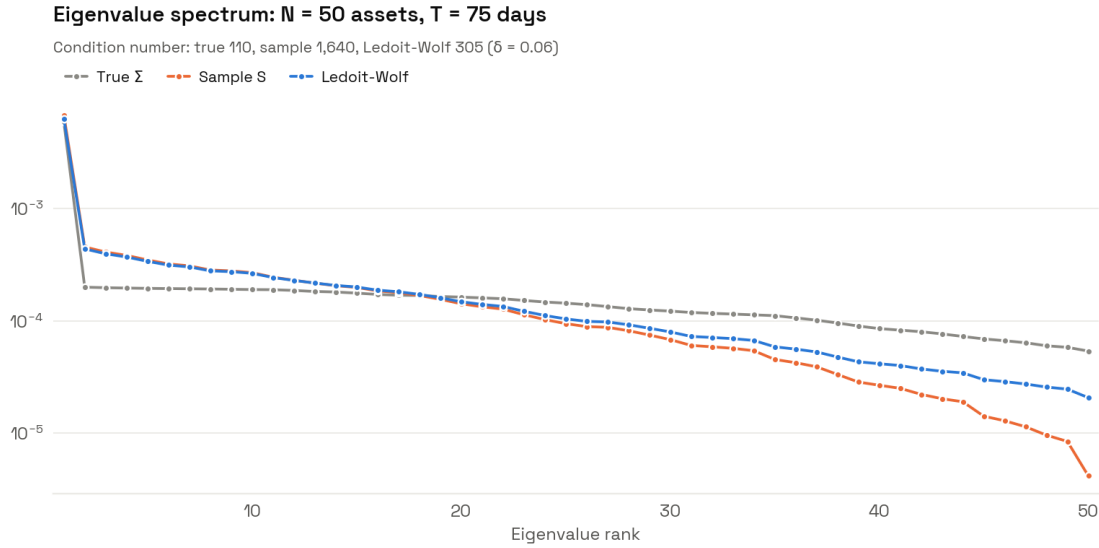


Figure 1. Eigenvalues of the true covariance, the sample covariance and Ledoit-Wolf (50 assets, 75 dates). Shrinkage lifts the small eigenvalues and divides the condition number by 5.

Definition 3.6 (EWMA). For a half-life h , let $\lambda = 2^{-1/h}$ and $\omega_s = \lambda^s / \sum_u \lambda^u$, where $s = 0$ is the most recent date. Then $\hat{\Sigma} = \sum_s \omega_s (R_{t-s} - \bar{R}_\omega)(R_{t-s} - \bar{R}_\omega)^\top$, with $\bar{R}_\omega = \sum_s \omega_s R_{t-s}$.

Proposition 3.7. Over an infinite history, the effective number of observations $(\sum_s \omega_s)^2 / \sum_s \omega_s^2$ equals $(1 + \lambda)/(1 - \lambda)$, about 182 for $h = 63$.

Proof. $\sum_s \lambda^s = \frac{1}{1-\lambda}$ and $\sum_s \lambda^{2s} = \frac{1}{1-\lambda^2}$, so the ratio is $\frac{1-\lambda^2}{(1-\lambda)^2} = \frac{1+\lambda}{1-\lambda}$. ■

In the code

Every rule accepts `cov_method="sample" | "ledoit_wolf" | "ewma"` or a function returns `-> ndarray`. Directly: `estimate_cov(R, "ledoit_wolf")`, `ledoit_wolf_cov(R)` which returns `(Sigma, delta)`, `ewma_cov(R, halflife=63)`.

In practice

No rule in the library uses expected returns. The standard error of an annualized mean is $\sigma/\sqrt{Y}$ over Y years, whatever the frequency ($\pm 5\%$ for $\sigma = 16\%$ and 10 years), whereas the error on a variance shrinks with the number of observations: covariances are estimated much better than means.

4 Risk contributions

Proposition 4.1 (Euler decomposition). *For any w with $\sigma_p(w) > 0$: $\frac{\partial \sigma_p}{\partial w} = \frac{\Sigma w}{\sigma_p}$ and $\sigma_p(w) = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i}(w)$.*

Proof. The gradient follows from differentiating $\sqrt{w^\top \Sigma w}$. Then $\sum_i w_i (\Sigma w)_i / \sigma_p = w^\top \Sigma w / \sigma_p = \sigma_p$. ■

Definition 4.2 (risk contributions). Marginal, total and relative risk contribution of asset i :

$$\text{MRC}_i = \frac{(\Sigma w)_i}{\sigma_p}, \quad \text{RC}_i = w_i \text{MRC}_i, \quad \frac{\text{RC}_i}{\sigma_p} = \frac{w_i (\Sigma w)_i}{w^\top \Sigma w}.$$

By proposition 4.1, the RC_i sum to σ_p and the relative contributions sum to 1.

Proposition 4.3. $\text{MRC}_i = \beta_{i,p} \sigma_p = \rho_{i,p} \sigma_i$, where $\beta_{i,p}$ and $\rho_{i,p}$ are the beta and the correlation of asset i with the portfolio.

Proof. $(\Sigma w)_i = \text{Cov}(R_i, w^\top R) = \text{Cov}(R_i, R_p)$. With $\beta_{i,p} = \text{Cov}(R_i, R_p) / \sigma_p^2$ and $\rho_{i,p} = \text{Cov}(R_i, R_p) / (\sigma_i \sigma_p)$, both identities follow. ■

An asset weighs on risk through its beta to the portfolio, not through its own volatility. If it is negatively correlated with the portfolio, its contribution is negative.

Example

60/40 equity/bond portfolio, $\sigma_E = 16\%$, $\sigma_B = 6\%$, $\rho = -0.2$:
 $\sigma_p^2 = 0.36 \cdot 0.0256 + 0.16 \cdot 0.0036 - 0.48 \cdot 0.2 \cdot 0.0096 = 0.00887$, so $\sigma_p = 9.4\%$. Then
 $(\Sigma w)_E = 0.6 \cdot 0.0256 - 0.4 \cdot 0.00192 = 0.01459$ and $\text{RC}_E / \sigma_p = 0.6 \cdot 0.01459 / 0.00887 = 98.7\%$: a 60/40 portfolio in capital is a 99/1 portfolio in risk.

In the code

`risk_contributions(w, cov)` returns the relative contributions;
`Backtester.risk_contributions()` gives them at every rebalancing.

5 Allocation rules

Definition 5.1 (allocation rule). An allocation rule is a function $\mathcal{A}$ that maps a window of returns $(R_{t-L}, \dots, R_{t-1})$ to a portfolio $w \in \Delta_N$. In the library, it is a `BaseEstimator:est.fit(window).weights_`.

5.1 Equal Weight (1/N)

Definition 5.2. $w_i = 1/N$ for all i .

Proposition 5.3. *If all assets have volatility σ and pairwise correlations ρ , then $\sigma_{1/N}^2 = \frac{\sigma^2}{N} (1 + (N-1)\rho) \rightarrow \rho \sigma^2$ as $N \rightarrow \infty$.*

Proof. $w^\top \Sigma w = \frac{1}{N^2} (N\sigma^2 + N(N-1)\rho\sigma^2)$. ■

Diversification removes the risk specific to each asset but not the common risk $\rho\sigma^2$. $1/N$ estimates nothing and so makes no estimation error: it is the reference benchmark, which none of the 14 optimization models tested by [DeMiguel et al. \(2009\)](#) significantly beats out of sample.

In the code
`EqualWeight()`

5.2 Inverse volatility

Definition 5.4. $w_i = \sigma_i^{-k} / \sum_j \sigma_j^{-k}$, with $k = 1$ (inverse volatility) or $k = 2$ (inverse variance).

Proposition 5.5. (i) *If all correlations are equal, inverse volatility equalizes risk contributions: it is the ERC portfolio (section 5.4).*

(ii) *If Σ is diagonal, inverse variance is the minimum variance portfolio.*

Proof. (i) With $w_i = c/\sigma_i$ and $\rho_{ij} = \rho$ for $i \neq j$: $(\Sigma w)_i = \sum_j \rho_{ij} \sigma_i \sigma_j c/\sigma_j = c \sigma_i (1 + (N-1)\rho)$, so $w_i(\Sigma w)_i = c^2(1 + (N-1)\rho)$ does not depend on i . (ii) The minimum variance portfolio is $w \propto \Sigma^{-1} \mathbf{1}$ and, for a diagonal Σ , $\Sigma^{-1} \mathbf{1} = (\sigma_i^{-2})_i$. ■

In the code
`InverseVolatility(exponent=1); exponent=2 for inverse variance.`

5.3 Risk budgeting

Definition 5.6. Let budgets $b_i > 0$ sum to 1. A risk budgeting portfolio is a $w \in \Delta_N$ such that $w_i(\Sigma w)_i / (w^\top \Sigma w) = b_i$ for all i : asset i carries the share b_i of the risk.

Theorem 5.7 ([Spinu, 2013](#)). *Let $\Sigma \succ 0$ and $f(y) = \frac{1}{2} y^\top \Sigma y - \sum_i b_i \ln y_i$ on $\mathbb{R}_{++}^N$. Then f has a unique minimizer y^* , and $w^* = y^* / \mathbf{1}^\top y^*$ is the unique risk budgeting portfolio.*

Proof. *Existence.* $f \rightarrow +\infty$ when some $y_i \rightarrow 0^+$ (log term) and when $\|y\| \rightarrow \infty$ (the quadratic term, bounded below by $\frac{1}{2} \lambda_{\min}(\Sigma) \|y\|^2$, dominates). The minimum is therefore attained in the interior of $\mathbb{R}_{++}^N$. *Uniqueness.* $\nabla^2 f = \Sigma + \text{diag}(b_i/y_i^2) \succ 0$: f is strictly convex. *Budgets.* $\partial_i f = (\Sigma y)_i - b_i/y_i = 0$ gives $y_i^*(\Sigma y^*)_i = b_i$. Since $w_i(\Sigma w)_i$ is homogeneous of degree 2, the relative contributions of w^* equal $b_i / \sum_j b_j = b_i$. *Uniqueness of the portfolio.* If w qualifies, $y = w / \sqrt{w^\top \Sigma w}$ satisfies $y_i(\Sigma y)_i = b_i$, so $\nabla f(y) = 0$, so $y = y^*$. ■

Proposition 5.8 (coordinate update). *With $(y_j)_{j \neq i}$ fixed, the minimizer of f in $y_i > 0$ is*

$$y_i = \frac{-c_i + \sqrt{c_i^2 + 4\Sigma_{ii}b_i}}{2\Sigma_{ii}}, \quad c_i = \sum_{j \neq i} \Sigma_{ij}y_j.$$

Proof. $\partial_i f = 0$ reads $\Sigma_{ii}y_i^2 + c_iy_i - b_i = 0$. The product of the roots, $-b_i/\Sigma_{ii}$, is negative: exactly one root is positive, and it is this one. ■

Repeating this update over $i = 1, \dots, N$ until it stabilizes (cyclical coordinate descent) converges to y^* (admitted, [Tseng, 2001](#)).

Algorithm 1: Risk budgeting by coordinate descent (**RiskParity**)

Input: $\Sigma \succ 0$, budgets b , tolerance $\varepsilon = 10^{-10}$
Output: weights w

```

1  $y_i \leftarrow b_i/\sigma_i; \quad y \leftarrow y/\sqrt{y^\top \Sigma y}; \quad s \leftarrow \Sigma y;$ 
2 repeat
3    $y^{\text{old}} \leftarrow y;$ 
4   for  $i \leftarrow 1$  to  $N$  do
5      $c \leftarrow s_i - \Sigma_{ii} y_i; \quad y'_i \leftarrow (-c + \sqrt{c^2 + 4\Sigma_{ii} b_i})/(2\Sigma_{ii});$ 
6      $s \leftarrow s + \Sigma_{:,i}(y'_i - y_i); \quad y_i \leftarrow y'_i; \quad \# O(N) \text{ cost, no inversion}$ 
7 until  $\|y - y^{\text{old}}\| \leq \varepsilon \|y\|;$ 
8 return  $w = y/\mathbf{1}^\top y$ 
```

In the code

`RiskParity(budgets={"SPY": 1, "TLT": 2, ...})`, one strictly positive budget per asset (normalized automatically); `n_iter_` gives the number of sweeps.

5.4 Equal Risk Contribution (ERC)

Definition 5.9. ERC is risk budgeting with $b_i = 1/N$: every asset carries the same share of the risk.

Theorem 5.10 (Maillard et al., 2010). *If σ_{MV} is the volatility of the long-only minimum variance portfolio, then $\sigma_{\text{MV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$.*

Proof. The first inequality holds because minimum variance minimizes σ_p over the whole of Δ_N . For the second, σ_p is convex (a norm of $\Sigma^{1/2}w$), hence lies above its tangent at $w = w_{\text{ERC}}$. With $\nabla \sigma_p(w)^\top w = \sigma_p(w)$ (proposition 4.1):

$$\sigma_{1/N} \geq \sigma_{\text{ERC}} + \nabla \sigma_p(w)^\top \left(\frac{1}{N} - w\right) = \frac{1}{N} \sum_i \partial_i \sigma_p(w).$$

At ERC, $w_i \partial_i \sigma_p = \sigma_{\text{ERC}}/N$, hence $\sigma_{1/N} \geq \frac{\sigma_{\text{ERC}}}{N^2} \sum_i \frac{1}{w_i} \geq \sigma_{\text{ERC}}$, since $\sum_i 1/w_i \geq N^2$ on Δ_N (arithmetic-harmonic mean inequality). ■

Proposition 5.11. *If all assets have the same Sharpe ratio and equal correlations, ERC is the maximum Sharpe portfolio.*

Proof. The maximum Sharpe portfolio is $w \propto \Sigma^{-1}\mu$. With $\mu_i = s\sigma_i$, $\Sigma = DCD$, $D = \text{diag}(\sigma_i)$ and $C = (1 - \rho)I + \rho\mathbf{1}\mathbf{1}^\top$: since $C\mathbf{1} = (1 + (N - 1)\rho)\mathbf{1}$, we get $\Sigma^{-1}\mu = sD^{-1}C^{-1}\mathbf{1} \propto D^{-1}\mathbf{1}$. This is inverse volatility, equal to ERC by proposition 5.5. ■

In the code

`ERC(cov_method="ledoit_wolf")`

5.5 Hierarchical Risk Parity (HRP)

HRP (López de Prado, 2016) groups assets that look alike, then splits capital between groups before splitting it within each group. It never inverts Σ .

Definition 5.12 (correlation distance). $d_{ij} = \sqrt{\frac{1}{2}(1 - \rho_{ij})} \in [0, 1]$.

Proposition 5.13. d is a metric, equal to zero if and only if $\rho_{ij} = 1$.

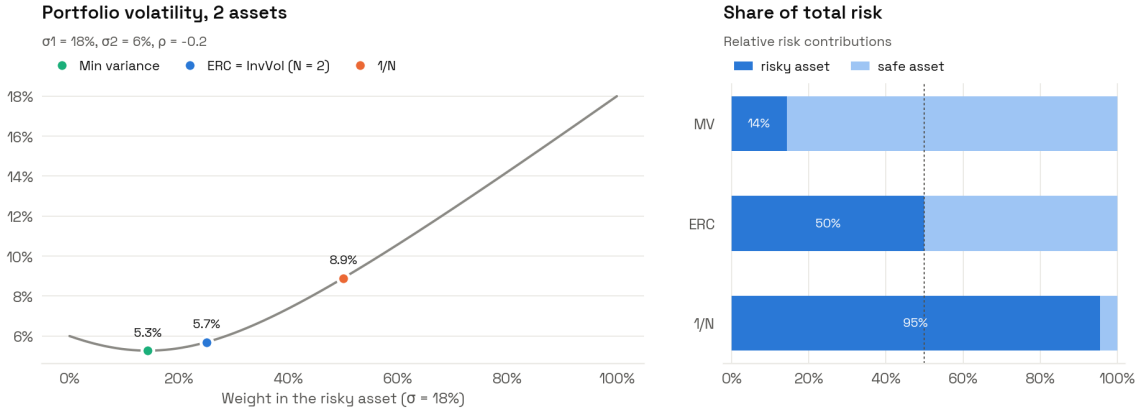


Figure 2. Two assets ($\sigma_1 = 18\%$, $\sigma_2 = 6\%$, $\rho = -0.2$). ERC has a volatility close to the minimum (5.7% versus 5.3%) but splits risk equally, whereas 1/N puts 95% of it on the risky asset.

Proof. Let z_i be the series of asset i , demeaned and scaled to unit norm. Then $\rho_{ij} = z_i^\top z_j$ and $\|z_i - z_j\|^2 = 2 - 2\rho_{ij}$, so $d_{ij} = \frac{1}{2}\|z_i - z_j\|$ inherits the properties of the Euclidean norm. ■

- Definition 5.14** (HRP). 1. *Clustering*: hierarchical clustering of the assets according to d (linkage method), which produces a binary tree.
2. *Quasi-diagonalization*: the assets are ordered following the leaves of the tree, which places correlated assets next to each other.
3. *Bisection*: start from $w = \mathbf{1}$ and recursively split each group into two sub-groups L and R (two halves of the ordered list, or the two branches of the tree). At each split,

$$\alpha = \frac{V_R}{V_L + V_R}, \quad w_L \leftarrow \alpha w_L, \quad w_R \leftarrow (1 - \alpha) w_R,$$

where $V_c = \tilde{w}_c^\top \Sigma_c \tilde{w}_c$ is the variance of the inverse variance portfolio of group c , with $\tilde{w}_c \propto \text{diag}(\Sigma_c)^{-1}$.

Each sub-group receives a share inversely proportional to its variance. The $\tilde{w}_c$ are only used to compute V_c : the final weights come from the successive splits, down to individual assets.

Proposition 5.15. (i) *HRP returns a portfolio in Δ_N and only inverts diagonals: it works even if Σ is singular.*

(ii) *If Σ is diagonal, HRP is the inverse variance portfolio.*

Proof. (i) Before a group c is split, all its assets carry the same weight m_c (they have gone through the same multiplications), with $m = 1$ for the whole universe. A split gives $m_L = \alpha m_c$ and $m_R = (1 - \alpha)m_c$ with $\alpha \in (0, 1)$ since $V_L, V_R > 0$. By induction on the tree, the sum of the final weights of the assets of c equals m_c : this is immediate for a single asset, and otherwise it equals $m_L + m_R = m_c$. For the whole universe the sum is 1, and all weights are positive. (ii) For a diagonal Σ , $V_c = 1/S_c$ with $S_c = \sum_{i \in c} \sigma_i^{-2}$, so $\alpha = S_L/(S_L + S_R)$: each group receives the inverse variance share of its assets, and by induction asset i receives $\sigma_i^{-2}/\sum_j \sigma_j^{-2}$. ■

In the code

HRP(linkage="single", split="bisection") reproduces the original paper;
 split="dendrogram" follows the branches of the tree instead of cutting the list into two halves.
 Attributes: order_ (leaf order), linkage_matrix_ (tree).

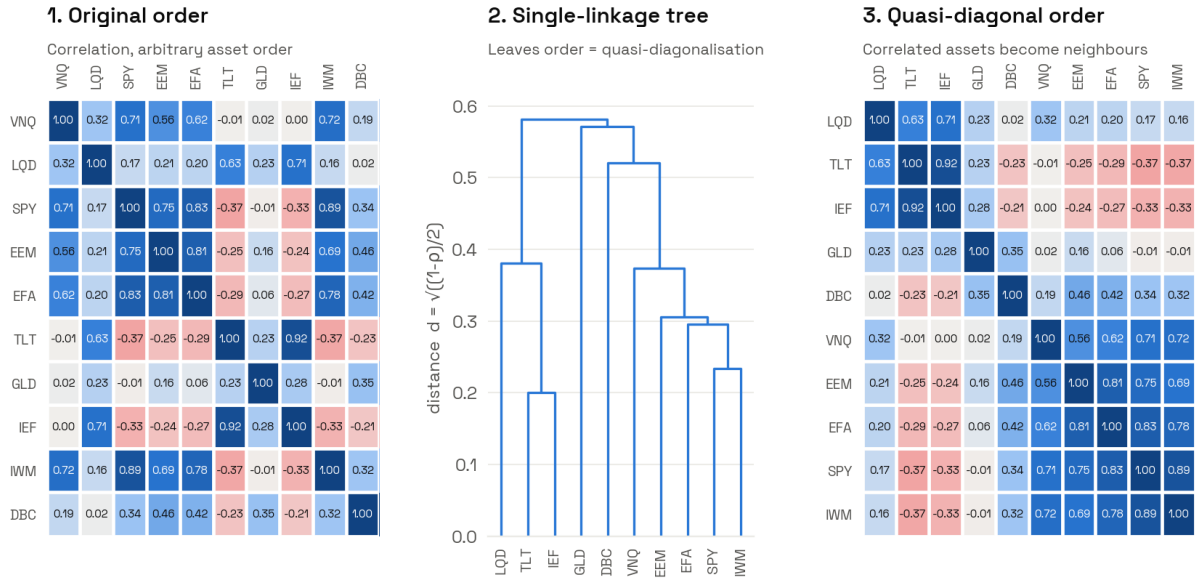


Figure 3. HRP steps on the simulated universe: correlations in an arbitrary order, tree, reordered correlations. Two blocks appear: rates and credit at the top, risky assets at the bottom.

In practice

The tree can change from one rebalancing to the next: HRP is less stable than ERC (turnover 2.5 times higher in the example). The "average" and "ward" methods give more stable trees than "single".

5.6 Summary

	1/N	InvVol	Risk budgeting, ERC	HRP
Uses σ_i	no	yes	yes	yes
Uses ρ_{ij}	no	no	yes	yes, through the tree
Inverts Σ	no	no	no	no
Computation	direct	direct	iterative, unique solution	direct
What is equalized	capital	$w_i \sigma_i$	$w_i (\Sigma w)_i$	variance across groups
Results	prop. 5.3	prop. 5.5	th. 5.7, 5.10	prop. 5.15

6 Walk-forward backtest

Definition 6.1 (walk-forward backtest). Given a rule $\mathcal{A}$, a window L and a holding period H , the rebalancing dates are $t_k = L + kH$. At each t_k we compute $w^{(k)} = \mathcal{A}(R_{t_k-L}, \dots, R_{t_k-1})$ (rolling window) or $\mathcal{A}(R_0, \dots, R_{t_k-1})$ (expanding window), then hold this portfolio until $t_{k+1} - 1$.

Proposition 6.2 (no look-ahead). The portfolio that earns the return R_t only depends on $R_0, \dots, R_{t-1}$.

Proof. If $t_k \leq t < t_{k+1}$, R_t is earned with $w^{(k)}$ drifted up to $t - 1$ (definition below). $w^{(k)}$ only depends on returns before $t_k \leq t$, and the drift only on $R_{t_k}, \dots, R_{t-1}$. ■

Walk-forward backtest (rolling window)

Weights fitted on returns up to $t-1$ are held on $[t, t+H]$, then the window rolls by H

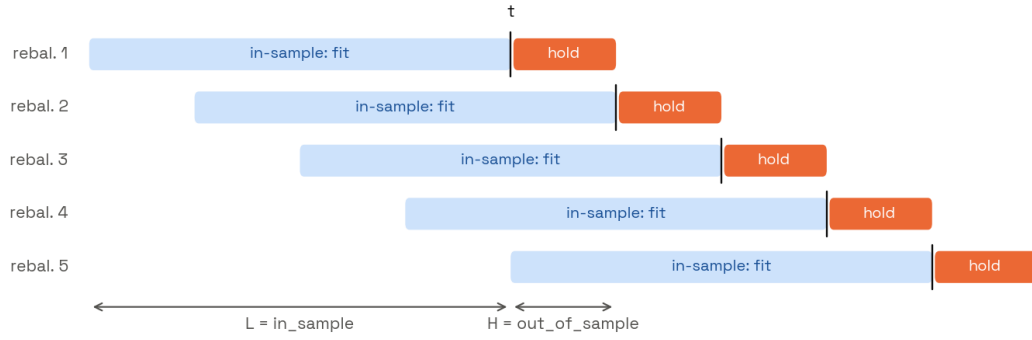


Figure 4. Estimation over $[t - L, t)$, holding over $[t, t + H)$, then a shift by H .

Definition 6.3 (weight drift). Between two rebalancings, the quantities held do not change. k days after the rebalancing ($V_0 = 1$):

$$V_k = \sum_i w_i \prod_{s=1}^k (1 + R_{i,s}), \quad R_{p,k} = \frac{V_k}{V_{k-1}} - 1.$$

Proposition 6.4. The weight of asset i after k days is $w_i^{(k)} = w_i \prod_{s \leq k} (1 + R_{i,s}) / V_k$, and $R_{p,k} = \sum_i w_i^{(k-1)} R_{i,k}$.

Proof. Position i is worth $w_i \prod_{s \leq k} (1 + R_{i,s})$ and its weight is this value divided by V_k . Then $\frac{V_k}{V_{k-1}} = \sum_i \frac{w_i \prod_{s \leq k} (1 + R_{i,s})}{V_{k-1}} (1 + R_{i,k}) = \sum_i w_i^{(k-1)} (1 + R_{i,k})$: this is proposition 2.3 applied to the drifted weights. ■

Definition 6.5 (turnover and costs). At rebalancing, turnover is $\tau = \sum_i |w_i^{\text{target}} - w_i^{\text{drifted}}|$, measured against the *drifted* weights and not against the previous target. With a proportional cost c , wealth is multiplied by $1 - c\tau$. Annualized turnover is the sum of the τ , excluding the initial build-up, divided by the number of years.

The annual cost is therefore about c times the annualized turnover: 120% turnover at 10 bps costs 12 bps a year.

Algorithm 2: Walk-forward backtest (Backtester.run)

Input: returns R , rule $\mathcal{A}$, L , H , cost c
Output: out-of-sample returns net of costs

```

1  $w^{\text{drifted}} \leftarrow 0$ ;  $t \leftarrow L$ ; # start in cash
2 while  $t < T$  do
3   eligible assets  $\leftarrow$  assets with no missing value over the window;
4    $w \leftarrow \mathcal{A}(\text{window})$ ;  $\tau \leftarrow \sum_i |w_i - w_i^{\text{drifted}}|$ ;
5   hold  $w$  over  $[t, t + H)$  with drift; wealth  $\times (1 - c\tau)$  at  $t$ ;
6    $w^{\text{drifted}} \leftarrow$  weights at period end;  $t \leftarrow t + H$ ;

```

In the code

```

bt = Backtester(R, ERC(), in_sample=252, out_of_sample=21,
transaction_cost_bps=5); bt.run() returns the net returns. Attributes weights_, turnover_,
gross_returns_; methods annualized_turnover() and risk_contributions(). Options:
window="expanding", drift=False (free daily rebalancing to target). run_backtests compares
several rules on the same grid.

```

7 Performance metrics

The functions of `portopt.metric` take a series of simple returns (`pd.Series`) or a table with one column per strategy (`pd.DataFrame`).

Definition 7.1 (total return, CAGR). $TR = \prod_{t=1}^T (1 + R_t) - 1$ and $CAGR = (1 + TR)^{P/T} - 1$: the constant annual rate leading to the same final wealth.

Proposition 7.2 (volatility drag). *The growth rate $g = \ln(1 + CAGR)$ satisfies, to second order, $g \approx \mu_{\text{ann}} - \frac{1}{2}\sigma_{\text{ann}}^2$.*

Proof. $g = \frac{P}{T} \sum_t \ln(1 + R_t)$ and $\ln(1 + R) = R - \frac{R^2}{2} + O(R^3)$. Since $\frac{1}{T} \sum_t R_t^2 = \hat{\sigma}^2 + \bar{R}^2 \approx \hat{\sigma}^2$ for daily data, $g \approx P\bar{R} - \frac{P}{2}\hat{\sigma}^2 = \mu_{\text{ann}} - \frac{1}{2}\sigma_{\text{ann}}^2$. ■

At equal mean, the less volatile strategy ends up richer. In the example, 1/N has $\mu_{\text{ann}} = 7.36\%$ and $\sigma_{\text{ann}} = 11.42\%$: the formula gives $g = 6.71\%$, versus 6.70% observed.

Definition 7.3 (Sharpe ratio). With an annual risk-free rate r_f converted per period, $SR = \sqrt{P} (\bar{R} - r_f) / \hat{\sigma}$.

Proposition 7.4. *The Sharpe ratio is invariant to leverage: multiplying excess returns by $k > 0$ leaves it unchanged. Under the assumptions of proposition 2.6, the annual Sharpe ratio is $\sqrt{P}$ times the per-period Sharpe ratio.*

Proof. The mean and standard deviation of kX are k times those of X . Annually, the ratio becomes $P\mu/(\sqrt{P}\sigma) = \sqrt{P}\mu/\sigma$. ■

Definition 7.5 (Sortino ratio). For a per-period threshold τ ,

$$\text{Sortino} = \sqrt{P} \frac{\bar{R} - \tau}{DD_\tau}, \quad DD_\tau = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t - \tau, 0)^2}.$$

The mean of squares runs over *all* dates, gains counting as zero. The Sortino ratio is the mean return per unit of downside risk.

Proposition 7.6. (i) For $\tau = 0$, the Sortino ratio is invariant to leverage.

(ii) If $R \sim \mathcal{N}(\mu, \sigma^2)$, $\tau = 0$ and $m = \mu/\sigma$:

$$\frac{\text{Sortino}}{SR} = \frac{1}{\sqrt{(1 + m^2)\Phi(-m) - m\varphi(m)}} = \sqrt{2} \left(1 + \sqrt{2/\pi} m \right) + O(m^2).$$

Proof. (i) For $k > 0$, $\min(kR, 0) = k \min(R, 0)$: numerator and denominator are multiplied by k . (ii) The ratio equals σ/DD_0 . With $R = \sigma(m + Z)$ and $Z \sim \mathcal{N}(0, 1)$, $DD_0^2 = \sigma^2 \mathbb{E}[(m + Z)^2 \mathbf{1}_{Z < -m}]$. The identities $\mathbb{E}[Z \mathbf{1}_{Z < a}] = -\varphi(a)$ and $\mathbb{E}[Z^2 \mathbf{1}_{Z < a}] = \Phi(a) - a\varphi(a)$, taken at $a = -m$, give $m^2\Phi(-m) - 2m\varphi(m) + \Phi(-m) + m\varphi(m)$, hence the formula. For the expansion, $\Phi(-m) = \frac{1}{2} - \varphi(0)m + O(m^2)$ and $\varphi(m) = \varphi(0) + O(m^2)$ with $\varphi(0) = 1/\sqrt{2\pi}$, so the radicand equals $\frac{1}{2} - 2\varphi(0)m + O(m^2)$. ■

For a symmetric distribution, the Sortino ratio is therefore about $\sqrt{2} \approx 1.41$ times the Sharpe ratio when the mean is zero, a bit more when it is positive. In the example (daily Sharpe ratio of 0.04 to 0.055), a Gaussian would give 1.46 to 1.48; we observe 1.44 to 1.51. The Sortino ratio then adds almost nothing to the Sharpe ratio. It becomes informative for clearly asymmetric strategies.

Definition 7.7 (Calmar, hit ratio). $\text{Calmar} = \text{CAGR}/|\text{MaxDD}|$ (MaxDD is defined in section 8) and $\text{Hit} = \frac{1}{T} \sum_t \mathbf{1}_{R_t > 0}$.

In the code

`cagr`, `annualized_mean`, `annualized_volatility`, `sharpe_ratio(r, rf=0.0)` (annual r_f), `sortino_ratio(r, target=0.0)` (*per-period* threshold), `calmar_ratio`, `hit_ratio`.
`summary(rets)` gathers them in a table.

8 Risk metrics

Definition 8.1 (drawdown). With wealth $W_t = \prod_{s \leq t} (1 + R_s)$ and $W_0 = 1$:

$$DD_t = \frac{W_t}{\max(1, \max_{s \leq t} W_s)} - 1, \quad \text{MaxDD} = \min_t DD_t.$$

The maximum time under water is the longest run of consecutive dates with $DD_t < 0$.

Initial capital counts as a peak: losing 10% on the first day gives a drawdown of -10% .

Proposition 8.2. *After a loss $\ell \in (0, 1)$, a gain of $\ell/(1 - \ell)$ is needed to get back to the initial level.*

Proof. $(1 - \ell)(1 + g) = 1 \iff g = \ell/(1 - \ell)$. So -20% requires $+25\%$, and -50% requires $+100\%$. ■



Figure 5. Peak, trough, recovery and time under water of the largest drawdown of the 1/N portfolio.

MaxDD mechanically worsens with the length of the observation period: for a Brownian motion without drift, its expectation grows like $\sigma\sqrt{T}$ (admitted, [Magdon-Ismail et al., 2004](#)). MaxDD values are therefore only compared over the same period.

Definition 8.3 (historical VaR and CVaR). With q_α the empirical quantile of order α of returns: $\text{VaR}_\alpha = -q_\alpha$ and $\text{CVaR}_\alpha = -\mathbb{E}[R \mid R \leq q_\alpha]$, expressed as positive losses. CVaR is the average loss over the worst fraction α of dates.

Proposition 8.4. (i) $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$.

(ii) *VaR is not subadditive: one can have $\text{VaR}(A + B) > \text{VaR}(A) + \text{VaR}(B)$.*

(iii) *CVaR is subadditive; it is a coherent risk measure (admitted, Artzner et al., 1999).*

Proof. (i) On the event $\{R \leq q_\alpha\}$, $-R \geq -q_\alpha$; take the conditional expectation. (ii) Two independent bonds each lose 100 with probability 4%, 0 otherwise. For each, $\text{VaR}_{95\%} = 0$. For their sum, the loss is 0, 100 or 200 with probabilities 92.16%, 7.68% and 0.16%: $\text{VaR}_{95\%} = 100 > 0 + 0$. ■

In this example, $\text{CVaR}_{95\%} = 80$ for each bond and $103.2 \leq 160$ for the sum: CVaR does reward diversification.

Daily return distribution and tail risk

1/N portfolio, excess kurtosis 14.1: the normal curve underestimates the tail (shaded)

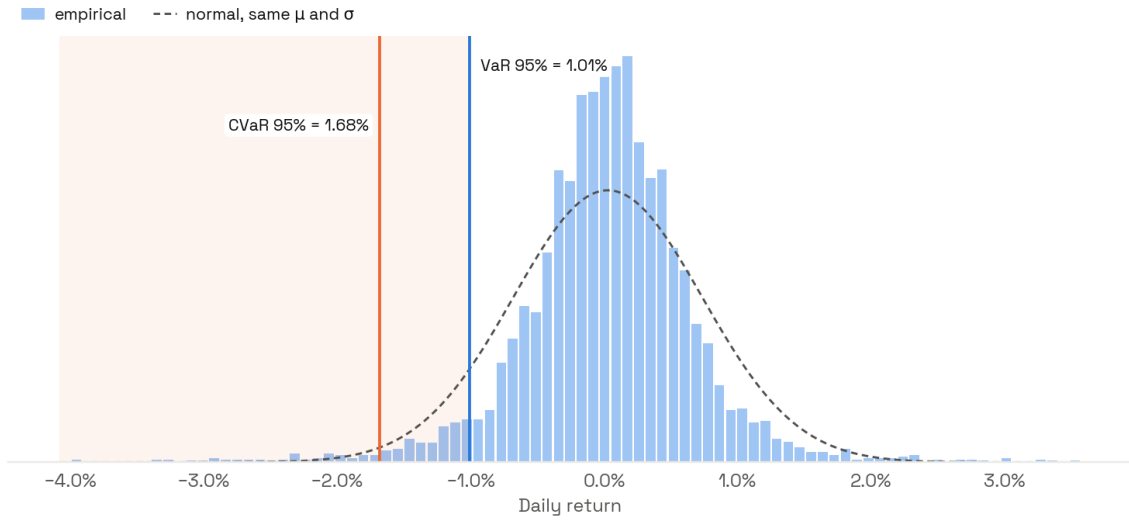


Figure 6. Daily returns of 1/N, 95% VaR and CVaR. The normal distribution with the same mean and variance underestimates the left tail.

Definition 8.5 (skewness and kurtosis). $\gamma_3 = \mathbb{E}[(R - \mu)^3]/\sigma^3$ and $\gamma_4 = \mathbb{E}[(R - \mu)^4]/\sigma^4$; the excess kurtosis $\gamma_4 - 3$ is zero for a normal distribution.

$\gamma_3 < 0$ signals rare but large losses, $\gamma_4 - 3 > 0$ tails fatter than a normal distribution. These moments depend on a few extreme days: check them over subperiods.

In the code

```
drawdown_series, max_drawdown, max_drawdown_duration, value_at_risk(r, alpha=0.05),
conditional_value_at_risk, skewness, excess_kurtosis.
```

9 Statistical significance

A backtest is a single path: a performance gap only matters if it exceeds the estimation noise.

Proposition 9.1 (standard error of the Sharpe ratio). *For independent normal returns with per-period Sharpe ratio SR , the Sharpe ratio estimated over T periods satisfies $\text{Var}(\widehat{\text{SR}}) \approx (1 + \frac{1}{2}\text{SR}^2)/T$. Annually, over Y years, $\text{se}(\widehat{\text{SR}}_{\text{ann}}) \approx \sqrt{(1 + \frac{1}{2}\text{SR}^2)/Y} \approx 1/\sqrt{Y}$.*

Proof. For a normal distribution, $\hat{\mu}$ and $\hat{\sigma}^2$ are independent, with $\text{Var}(\hat{\mu}) = \sigma^2/T$ and $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/T$. For $g(\mu, \sigma^2) = \mu/\sqrt{\sigma^2}$, $\partial_\mu g = 1/\sigma$ and $\partial_{\sigma^2} g = -\mu/(2\sigma^3)$. The delta method gives

$\text{Var}(\widehat{\text{SR}}) \approx \frac{1}{T} + \frac{\mu^2}{4\sigma^6} \cdot \frac{2\sigma^4}{T} = \frac{1+\text{SR}^2/2}{T}$. Annually, $\widehat{\text{SR}}_{\text{ann}} = \sqrt{P} \widehat{\text{SR}}$ and $T = PY$. ■

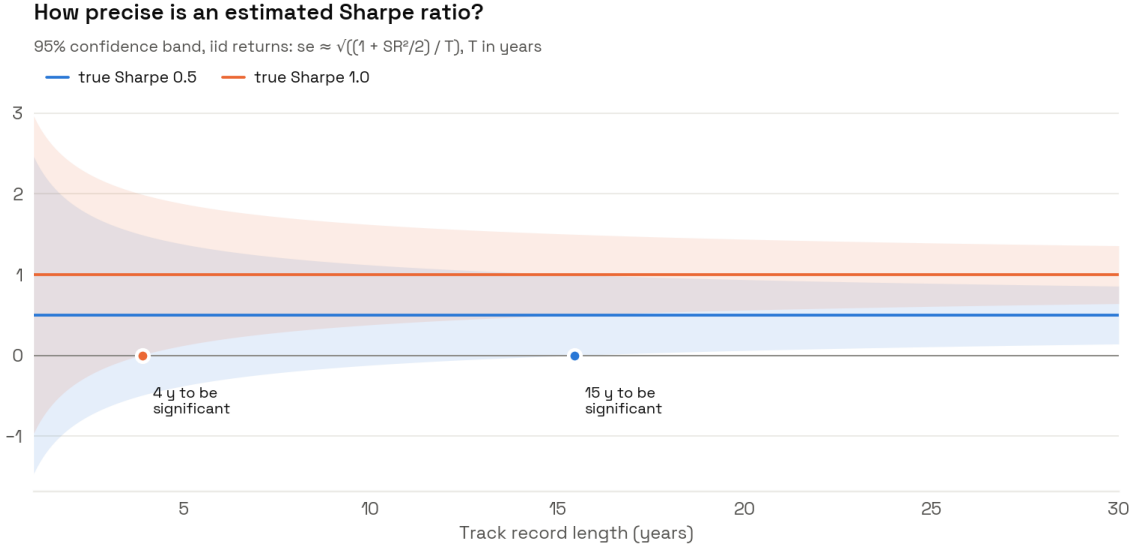


Figure 7. 95% confidence band of the estimated Sharpe ratio as a function of track record length.

A Sharpe ratio of 0.5 is only significantly positive after $(1.96/0.5)^2 \approx 15$ years. In the example, Sharpe ratios range from 0.64 to 0.88 over 14 years, with a standard error of about 0.27: the gaps between strategies are not significant.

Definition 9.2 (Probabilistic Sharpe Ratio). With $\widehat{\text{SR}}$ and a threshold SR^* expressed per period, and γ_3, γ_4 the skewness and (non-excess) kurtosis of returns,

$$\text{PSR}(\text{SR}^*) = \Phi \left(\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{T-1}}{\sqrt{1 - \gamma_3\widehat{\text{SR}} + \frac{\gamma_4-1}{4}\widehat{\text{SR}}^2}} \right).$$

It is the probability that the true Sharpe ratio exceeds SR^* , accounting for skewness and fat tails (admitted, [Bailey and López de Prado, 2012](#)). For $\gamma_3 = 0$ and $\gamma_4 = 3$, it reduces to proposition 9.1.

Definition 9.3 (Deflated Sharpe Ratio). If K variants were tested and the best one kept, the DSR is the PSR computed with the threshold

$$\text{SR}_K^* = \sqrt{\text{Var}(\widehat{\text{SR}}_k)} \left[(1 - \gamma_E) \Phi^{-1}(1 - \frac{1}{K}) + \gamma_E \Phi^{-1}(1 - \frac{1}{K\epsilon}) \right],$$

which approximates the expected maximum of K zero Sharpe ratios, with $\gamma_E \approx 0.5772$ (admitted, [Bailey and López de Prado, 2014](#)).

For $K = 100$, the bracket is about 2.53: an observed Sharpe ratio of at least 2.5 standard errors is needed to rule out the effect of picking the best trial.

In the code

```
probabilistic_sharpe_ratio(r, sr_benchmark=0.0); the threshold is given annualized and
converted internally. summary reports PSR(SR>0).
```

10 Full example

`python examples/run_backtest.py --synthetic: 10 simulated ETFs, 252-day window, re-balancing every 21 days, 5 bps of costs, December 2010 to June 2024.`

	1/N	InvVol	RB bond tilt	ERC	ERC LW	HRP
CAGR	6.9%	6.6%	6.1%	6.2%	6.2%	6.1%
Volatility	11.4%	8.9%	7.7%	8.2%	8.4%	7.0%
Sharpe	0.64	0.76	0.81	0.77	0.76	0.88
Max DD	−27.3%	−20.5%	−18.0%	−19.0%	−19.5%	−15.0%
CVaR 95% (daily)	1.7%	1.3%	1.1%	1.2%	1.2%	1.0%
Annual turnover	38%	41%	45%	49%	48%	120%

Table 1. Out-of-sample metrics, net of costs. RB bond tilt doubles the risk budget of TLT, IEF and LQD; ERC LW uses Ledoit-Wolf.

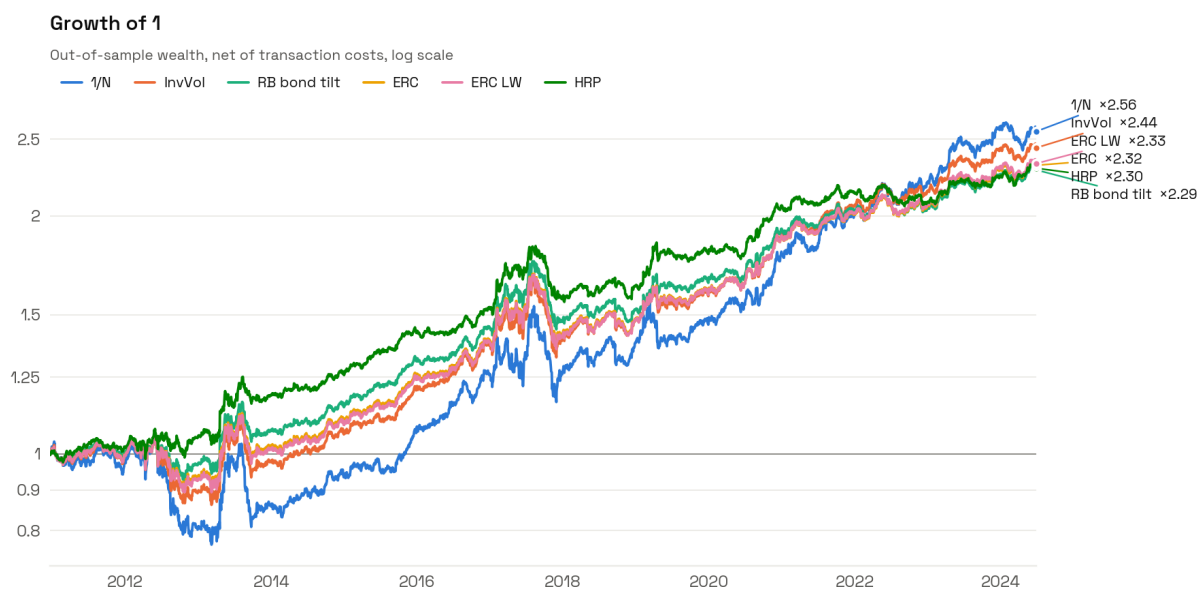


Figure 8. Out-of-sample growth of 1, net of costs.

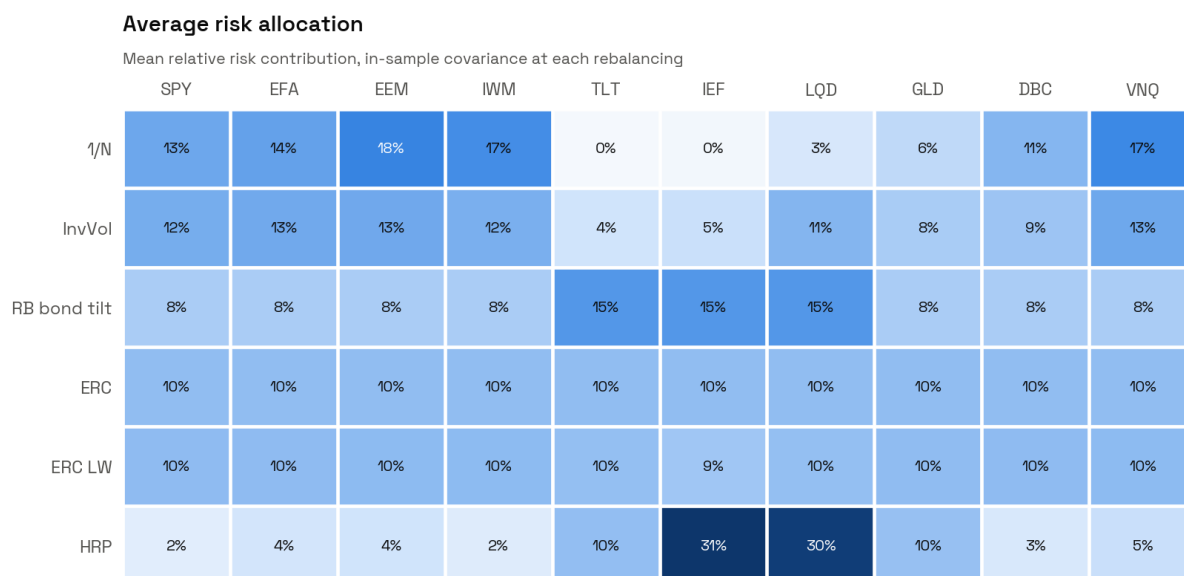


Figure 9. Average relative risk contributions. 1/N puts 79% of the risk on equity-like assets, ERC equalizes it, HRP concentrates 71% of it on the three bond ETFs.

Reading.

1. Compared with $1/N$, the risk-based rules cut volatility by 20 to 40% and drawdown by 25 to 45%. $1/N$ keeps the best CAGR because it is more exposed to risky assets.
2. The Sharpe ratio gaps between rules are smaller than one standard error (proposition 9.1): this backtest cannot rank them.
3. With 10 assets and 252 dates, Ledoit-Wolf shrinkage is small (median δ^* of 4%) and changes the result little.
4. HRP has the highest turnover, a consequence of the instability of the tree.

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